

MAKERERE UNIVERSITY BUSINESS SCHOOL

FACULTY OF COMPUTING AND INFORMATICS

Department of Computer Science and Engineering
Academic Year 2025/2026 — Semester II

SEMESTER PROJECT GUIDE

Programme	Bachelor of Business Computing — Year Three
Course Name	Web Server Administration
Course Code	BUC3219
Credit Units	4
Facilitators	Ms. Hajara Namuwaya Mr. Samuel Ssendi Mr. Nyero A

Specific Instructions

- Work in your class groups. Every member must contribute and must be prepared to answer questions independently during presentations, as respondents will be selected at random.
- Submit both the PDF report and any required files (Part I: screenshots and documentation; Part II: GitHub repository link) via MUBSEP. The PDF report must have a cover page listing all group members with their student and registration numbers.
- Show all necessary processes and, where applicable, justify the decisions and choices made during configuration.
- Document every stage of your work with annotated screenshots and written explanations. Include any challenges or errors encountered and describe how they were resolved.
- Avoid copying and pasting from external sources, copying between groups, or any other form of plagiarism and academic dishonesty.
- All groups are required to check for AI-generated content and plagiarism via Turnitin before uploading the final project report to MUBSEP. The plagiarism level must be below 20%.
- The project report must not be less than 10 pages and must not exceed 20 pages, excluding cover page, references, and screenshots.
- Each member must complete a Self and Peer Evaluation of Group Project via Katusome and submit it before the deadline for both Part I and Part II.
- All university regulations related to examination malpractices apply. Culprits will be penalised accordingly.

1. Project Description

This semester project gives student groups hands-on experience running a web hosting operation from the ground up. Rather than working in a simulated environment, each group acquires a real Virtual Private Server and a registered domain name, then builds, secures, and operates a production-grade hosting setup. The project runs in two phases. In Phase I, groups configure everything from the command line with no graphical control panel. In Phase II, they

introduce CWP (Control Web Panel) as a management layer on top of the infrastructure they have already built by hand.

The progression is intentional. The skills gained in each phase feed directly into the next. Students who skip ahead and rely on CWP without completing Phase I will not have the foundation needed to configure it correctly, troubleshoot it, or explain what it is doing.

2. Objective

The project gives students direct, practical experience in deploying, configuring, managing, and securing web servers in a virtualised environment. It is designed to reinforce what is covered in class and develop the technical competencies required of a web server administrator in a professional setting.

3. Learning Outcomes

On completing this project, students should be able to:

- Deploy and configure a Linux-based VPS from scratch.
- Install and manage core web services including Apache, Nginx, MySQL, PHP, and Postfix.
- Configure SSH, FTP, and SMTP services correctly and securely.
- Host multiple websites on a single server and manage domain and subdomain configurations.
- Apply firewall rules, SSL/TLS certificates, and server hardening measures.
- Manage DNS records and understand how they map to server infrastructure.
- Set up and use a mail server to send and receive professional email.
- Install and administer CWP as a control panel for a multi-user hosting environment.
- Deploy applications from GitHub directly to a server.
- Read and interpret server logs and monitor server performance.

4. Group Setup and Company Concept

Each group (maximum 10 members) will operate as a startup web hosting company throughout the project. Before any technical work begins, the group must agree on the following:

- A unique company name, one word or a short phrase. This name is used to register the group domain.
- A short company tagline/slogan describing the hosting service the company offers.
- Assigned roles within the group, for example: System Administrator, Network Administrator, Security Officer, Documentation Lead.

The group registers one domain name based on their company name (for example, ssendi.dev or hajarah.co.ug). The main domain hosts the group company website. Each group member gets a personal subdomain, such as hellen.ssendi.dev or festo.hajarah.co.ug, which they configure and manage individually throughout both project phases.

Every group member is required to work on their own subdomain site. Both a group report and an individual report are required for each submission, as detailed in Section 8.

5. VPS and Domain Requirements

Groups may choose any reputable VPS provider. The table below shows the minimum specifications required. Groups planning ahead for Phase II should note that CWP requires more resources than a bare Linux setup.

Operating System	Ubuntu 22.04 LTS or Rocky Linux 9 (recommended). CentOS Stream 9 is acceptable.
CPU	Minimum 2 vCores. 2 vCores also sufficient for Phase II with CWP.
RAM	Minimum 2 GB. 4 GB recommended for Phase II with CWP.
Storage	Minimum 20 GB NVMe SSD.
Domain	Any TLD is acceptable — .com, .net, .ug, and .co.ug are all fine.
DNS Management	Cloudflare (free) is recommended for DNS management.

Note: Groups on a 1 GB RAM VPS will need to upgrade before installing CWP in Phase II. CWP will not run stably below 2 GB RAM.

6. Project Deliverables

The project is divided into two parts. Part I contributes to Coursework Two. Part II forms the project component of the final examination.

Part I: Linux Web Server Setup — Without a Control Panel (Coursework Two)

Assessment: Coursework Two | Marks: 50

Submission Deadline: Sunday, 26 April 2026 at 1:00 PM via MUBSEP

Defence Presentations: 27 April to 1 May 2026 during class time

File Deployment: Download system files from the GitHub link posted on MUBSEP and upload to the server via FTP

In Part I, students configure a fully working Linux web server entirely from the command line. No graphical control panel is used at any point in this phase. Students who understand the underlying configuration will be far better placed to use and troubleshoot CWP in Phase II.

Task 1: VPS Acquisition and Initial Linux Setup [5 marks]

Set up the server environment and secure initial access.

1. Acquire a VPS and record the provider name, plan details, data centre location, and monthly cost.
2. Install the operating system and document the installation process.
3. Connect via SSH as root and confirm access.
4. Update all system packages to their latest versions.
5. Set a hostname that reflects the group company name.
6. Create a non-root administrative user with sudo privileges and disable direct root SSH login.

7. Configure SSH key-based authentication for the admin user and disable password-based login.
8. Set the correct timezone on the server.

Screenshots required: VPS dashboard showing plan details, successful SSH connection, hostname confirmation, and SSH key authentication in use.

Task 2: Firewall and Server Security [6 marks]

Harden the server before installing any web services.

1. Install and enable UFW.
2. Set the default policy: deny all incoming, allow all outgoing.
3. Open only the ports required for the planned services. Justify each open port in the report.
4. Implement at least three additional hardening measures of your choice — for example, fail2ban, disabling unused services, SSH configuration hardening, or automatic security updates. Explain each measure and why it matters.
5. Verify the full rule set is active and document it.
6. Test that blocked ports are unreachable and record the results.

In the report, include a table listing every open port, the service it supports, and the reason it is open.

Task 3: Web Server Installation and Virtual Hosts [7 marks]

Install and configure Apache and Nginx, then set up virtual hosts for all group domains and subdomains.

1. Install both Apache and Nginx on the server.
2. Document which web server mode you are using (Apache only, Nginx only, or Apache-behind-Nginx reverse proxy) and explain the choice.
3. Locate the main configuration files for both servers. In the report, annotate the key directives and explain what each one controls.
4. Create a virtual host for the group main domain (e.g., ssendi.dev), including the correct document root, server name, and www alias.
5. Create individual virtual host configurations for each group member's subdomain.
6. Set the correct file ownership and permissions for each document root. Explain in the report why these matter.
7. Deploy a company Site from the files downloaded from GitHub [Task 6].
8. Test all virtual host configurations and confirm every domain and subdomain loads correctly in a browser.

Screenshots required: configuration file contents with annotations in the report, and browser showing each domain and subdomain loading correctly.

Task 4: DNS Configuration [5 marks]

Connect the domain name to the server infrastructure.

1. Add the domain to Cloudflare (or the chosen DNS manager) and update the registrar nameservers.
2. Create an A record for the root domain pointing to the VPS IP address.
3. Create an A record for www pointing to the same IP.
4. Create A records for each group member's subdomain.

5. Create an MX record pointing to the mail server configured in Task 7.
6. Add an SPF TXT record for the domain.
7. Verify propagation of all records using command-line tools and document the results.
8. In the report, explain the purpose and structure of each DNS record type created (A, CNAME, MX, TXT, NS).

Task 5: SSL/TLS Certificates [5 marks]

Secure all hosted sites with HTTPS.

1. Install Certbot and the appropriate web server plugin.
2. Obtain a Let's Encrypt certificate for the main domain (including www).
3. Obtain certificates for each group member's subdomain.
4. Configure all virtual hosts to redirect HTTP to HTTPS automatically.
5. Verify all sites load over HTTPS with a valid certificate.
6. In the report, explain what TLS does, why HTTP alone is not sufficient, and how Let's Encrypt certificate renewal works.

Task 6: FTP Server Setup and File Deployment [5 marks]

Configure FTP and use it to deploy the provided web application files.

1. Install and configure vsftpd: disable anonymous access, enable local user login, chroot users to their directories, and set passive mode with a defined port range.
2. Create FTP accounts for each group member restricted to their respective web directories.
3. Open the required FTP ports in UFW, including the passive mode range.
4. Connect using FileZilla (or equivalent) and confirm file upload and download works for each account.
5. Download the system files from GitHub (link on MUBSEP) and use FTP to upload them to each member's site directory.
6. Confirm the deployed site is accessible in a browser via the correct subdomain.
7. In the report, explain the difference between FTP, SFTP, and FTPS, and the security implications of each.

Task 7: Mail Server Setup [7 marks]

Configure a mail server and create working professional email accounts.

1. Install Postfix and Dovecot on the server.
2. Configure Postfix to send and receive mail for the domain.
3. Configure Dovecot for IMAP access.
4. Create at least two professional email accounts, one per active group member, in the format firstname@yourdomain.com.
5. Add SPF DNS records for the domain.
6. Send an email from one group member's account to the other and from the server to an external address such as Gmail. Document the results.
7. Receive an external email sent to one of the professional addresses and document the result.
8. Access email via a webmail client such as Roundcube, or a mail client configured with IMAP.
9. In the report, explain the roles of Postfix, Dovecot, SPF, DKIM, and DMARC in a mail infrastructure.

Task 8: Database and phpMyAdmin [5 marks]

Install MySQL and provide a web-based management interface.

1. Install MySQL Server and run the secure installation script.
2. Create a separate database and database user for each group member's application. Grant each user only the privileges they need for their own database.
3. Install phpMyAdmin and configure it at a non-default URL path.
4. Verify each group member can log into phpMyAdmin and manage their own database.
5. Verify the deployed web application connects to its database and works correctly.
6. In the report, explain SQL injection, why dedicated database users matter, and how phpMyAdmin access should be restricted in a production environment.

Task 9: Log Analysis and Performance Monitoring [5 marks]

Read and interpret real server log data.

1. Locate the access and error log files for the web server and record their paths in the report.
2. Use command-line tools to find: the top 10 IP addresses accessing the server, the most frequently requested URLs, any 404 errors, and any 500 errors.
3. Install htop and take a screenshot showing CPU and memory usage.
4. Check disk usage across all mounted filesystems.
5. List all listening ports and the process behind each one.
6. Write a paragraph in the report interpreting what the log data reveals about the server's activity and health.

Part I: Marking Scheme

Task	Description	Marks
Task 1	VPS setup, SSH hardening, user accounts	5
Task 2	Firewall configuration and server hardening	6
Task 3	Apache/Nginx virtual hosts, document roots, permissions	7
Task 4	DNS records configuration and verification	5
Task 5	SSL/TLS certificates and HTTPS enforcement	5
Task 6	FTP setup and file deployment	5
Task 7	Mail server, professional email accounts, send/receive tests	7
Task 8	MySQL, database users, phpMyAdmin	5
Task 9	Log analysis and performance monitoring	5
TOTAL	Part I — Coursework Two	50

Defence presentations (27 April to 1 May): groups must demonstrate their live server, answer questions on any task, and explain configuration decisions. Undocumented work and tasks that cannot be demonstrated live will attract mark deductions.

Part II: CWP-Managed Hosting Environment (Final Examination Project)

Assessment: Final Examination — Project Component | Marks: 60 (60% of the final exam)

Submission Deadline: Sunday, 10 May 2026 at 1:00 PM via MUBSEP

Defence Presentations: Tuesday, 13 May 2026

File Deployment: Deploy system files directly from GitHub to the CWP-hosted environment

Part II builds on the infrastructure students configured by hand in Part I. The group installs CWP (Control Web Panel) on the server to simulate how a real hosting company manages multiple client accounts. Students will also deploy from GitHub for the first time, manage DNS and email through the panel, and harden a CWP-managed environment.

CWP is free and runs on CentOS Stream 9 or Rocky Linux 9. Groups using Ubuntu for Part I should review CWP's system requirements before beginning Part II and plan any OS migration or VPS re-provisioning in advance.

Task 1: CWP Installation and Initial Configuration [6 marks]

1. Research CWP's system requirements. Determine whether the current VPS meets them. Upgrade or re-provision as necessary and document what changed and why.
2. Prepare the server: confirm the OS is a supported version, the hostname is a fully qualified domain name, and the server is fully updated.
3. Install CWP. Document every step, including any errors encountered and how they were resolved.
4. Access the CWP admin panel and complete the post-installation configuration.
5. In the report, explain the difference between CWP (free) and cPanel (commercial). Identify at least five things CWP handles that were done manually in Part I.

Task 2: User Accounts and Hosting Packages [5 marks]

1. Create a hosting package in CWP with resource limits appropriate for student projects: disk space, bandwidth, number of databases, and email accounts.
2. Create a CWP user account for each group member, assigned to the package and linked to the member's subdomain.
3. Verify each member can log into the CWP user panel.
4. In the report, explain why resource limits and account isolation matter in a shared hosting environment.

Task 3: Web Server Configuration via CWP [6 marks]

1. Review the web server options in CWP (Apache only, Nginx only, Apache with Nginx as reverse proxy). Select the mode that fits the group's setup and justify the choice.
2. Add the group main domain and each member's subdomain as hosted accounts through CWP.
3. Set the document root and index file for each account.
4. Verify every domain and subdomain loads correctly.
5. Test an .htaccess redirect rule on one of the hosted sites and confirm it works.

Task 4: GitHub Deployment [8 marks]

1. Each group member forks or clones the project repository (link on MUBSEP) to their own GitHub account.
2. On the server, configure Git and set up SSH key authentication with GitHub.
3. For each member's account, clone the repository directly into the correct document root.
4. Verify the deployed application is accessible and working via the member's subdomain.
5. Demonstrate an update: make a change in the repository, pull it on the server, and verify the change is live. Document this with screenshots.
6. In the report, explain the advantages of Git-based deployment over FTP. Describe what a CI/CD pipeline is and how it extends this workflow.

Task 5: DNS and SSL via CWP [6 marks]

1. Use CWP's DNS Manager to review and update DNS zones for the domain. Confirm all required records are in place: A, MX, TXT/SPF.
2. Use CWP's AutoSSL to install Let's Encrypt certificates for all hosted accounts.
3. Verify all sites enforce HTTPS and certificates are valid.
4. In the report, compare managing DNS and SSL manually (Part I) with doing so through CWP. What does the panel handle automatically? What risks remain?

Task 6: Mail Server and Professional Email via CWP [9 marks]

1. Confirm the mail server components are enabled and correctly configured in CWP.
2. Create at least two professional email accounts through CWP's email manager, one per active group member.
3. Configure DKIM for the domain through CWP and add the DKIM DNS record.
4. Send an email between the two professional accounts and confirm delivery.
5. Send an email from the professional address to an external address such as Gmail and receive a reply.
6. Access email through CWP's webmail interface and take screenshots showing successful send and receive.
7. In the report, explain what DKIM is, how it differs from SPF, and why both are needed for reliable email delivery.

Task 7: Database and phpMyAdmin via CWP [6 marks]

1. Use CWP's MySQL Manager to create a database and user for each member's application.
2. Assign each user to their database with the correct privileges.
3. Access phpMyAdmin through CWP and confirm each member can manage their own database.
4. Confirm the deployed application connects to its database and works correctly.
5. In the report, compare CLI database management (Part I) with CWP's interface. Identify at least two risks in a shared hosting environment.

Task 8: Firewall and Security Hardening via CWP [7 marks]

1. Enable and configure CSF (ConfigServer Security and Firewall) through CWP.
2. Review the default CSF rule set and adjust it so only required services are accessible.
3. Enable Login Failure Detection in CSF. Demonstrate that repeated failed login attempts result in an IP block and document the test.
4. Compare CSF in CWP with the manual UFW/iptables approach used in Part I. Note similarities and differences.

- Run CWP's security audit and address at least two of its recommendations. Explain what was changed and why.

Task 9: Performance Monitoring and Log Analysis via CWP [7 marks]

- Use CWP's monitoring dashboard to observe CPU, RAM, and disk usage and take screenshots.
- Access per-account access and error logs through CWP. Identify at least three patterns or events and explain their significance.
- Compare CWP's log viewer with command-line log analysis (grep, awk, tail). Describe what each approach can do that the other cannot.
- Use CWP's process manager to check for any processes using unexpectedly high resources.

Part II: Marking Scheme

Task	Description	Marks
Task 1	CWP installation and initial configuration	6
Task 2	User accounts and hosting packages	5
Task 3	Web server configuration and virtual hosts via CWP	6
Task 4	GitHub deployment and update workflow	8
Task 5	DNS and SSL via CWP	6
Task 6	Mail server, professional email accounts, DKIM, send/receive tests	9
Task 7	Database management and phpMyAdmin via CWP	6
Task 8	CSF firewall and security hardening via CWP	7
Task 9	Performance monitoring and log analysis via CWP	7
TOTAL	Part II — Final Examination Project (60% of exam)	60

Defence presentations (13 May): each group demonstrates their live CWP environment, deploys an update via GitHub, and answers individual questions. All group members must be able to speak to any part of the project.

7. Assessment Summary

Component	Details	Weight
Coursework Two (Part I)	Tasks 1 to 9 and Defence Presentations (27 April to 1 May 2026)	50 marks
Final Exam (Part II Project)	Tasks 1 to 9 and Defence Presentation (13 May 2026)	60% of exam
Final Exam (Practical)	Written Examination (Practical)	40% of exam

The written examination covers all course outline topics from class and both project phases.

8. Report Structure and Submission Requirements

8.1 Group Report

One group report is required per phase. It must include:

- Cover page: company name and domain, group member names and student numbers, course details, and submission date.
- Company profile: company name, tagline, member roles, and the domain name purchased with a receipt or screenshot.
- VPS details: provider, plan, IP address, and operating system.
- Task sections in order (Tasks 1 to 9 for each phase). Each section must describe what was done, explain the key configuration choices made, and include annotated screenshots of the working result. Do not paste raw command lists without explanation, marks are awarded for understanding, not transcription.
- Turnitin plagiarism and AI detection report showing a score below 20%.
- Reflection: what the group found most difficult, what they would do differently, and what they took away from the experience.
- Live links: group main site URL and each member's subdomain URL.

8.2 Individual Member Report

Each group member submits their own individual report covering:

- Personal subdomain configuration: virtual host, document root, permissions, and SSL.
- FTP deployment steps (Part I) or GitHub deployment steps (Part II).
- Database and email account setup.
- Individual contribution to the group tasks.
- Live link to the personal subdomain site.

8.3 Submission Rules

Format	PDF only. Word documents will not be marked.
Platform	MUBSEP, submit both the group report and the individual report.
Screenshots	Must be clear, full-screen, and wherever possible show the system date and time.

Live sites	All hosted sites must be online at the time of submission and presentation.
Late submissions	Will not be accepted under any circumstances.
Academic honesty	All work must be the group's own. Makerere University examination rules apply in full. Plagiarism or shared work between groups will be referred for disciplinary action.
Plagiarism check	All groups must run a Turnitin check before submission. Plagiarism level must be below 20%. Reports submitted above this threshold will not be marked.
Page limit	The project report must not be less than 4 pages and must not exceed 6 pages, excluding the cover page, references, and screenshots.
Peer evaluation	Each member must complete the Self and Peer Evaluation of Group Project via Katusome and submit it before the deadline for both Part I and Part II.

9. Key Dates

Now to 20 April 2026	Part I: set up VPS and domain; complete Tasks 1 to 9. GitHub links will be posted on MUBSEP.
Sunday, 26 April 2026 at 1:00 PM	Part I submission deadline on MUBSEP.
27 April to 1 May 2026	Part I defence presentations during class time.
2 May to 9 May 2026	Part II: install CWP; complete Tasks 1 to 9; deploy from GitHub.
Sunday, 10 May 2026 at 1:00 PM	Part II submission deadline on MUBSEP.
Tuesday, 13 May 2026	Part II defence presentations.
TBA	Written examination (40% of final exam).

Makerere University examination rules and regulations apply. Students are expected to adhere to them throughout the project.